

Final Design Report

MM X Subsystem



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Declaration

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Chapter 1

Introduction

1.1 Problem Description

MMs are used in competitions and research to test autonomous robots' ability to navigate mazes efficiently. They support advancements in path-finding, sensor integration, and real-time control, making them a valuable platform for robotics education and innovation. The main objective of this project is to design a PCB for the MM power subsystem. The MM system requires a compact, efficient power subsystem capable of delivering regulated power and control signals to multiple components, including the motherboard, sensor, and processor. The PCB must comply with the predefined electrical interface standards of the pre-designed motherboard, supplying the necessary current and voltage under load to ensure success.

1.2 Scope and Limitations

The scope of this project is to design a power subsystem capable of supplying regulated voltages to the MM's processor and sensor modules while meeting integration requirements defined by the existing motherboard. Several key aspects of the scope are outlined below:

- **Hardware Design:** This involves conceptualizing and designing the electrical circuitry that makes up the power subsystem.
- **Component Selection:** Identifying and selecting appropriate components to meet the requirements of the MM robot, while considering constraints such as size, weight, cost, and efficiency.
- **Testing and Validation:** Performing comprehensive testing and validation procedures to verify the functionality and performance of the power subsystem.

Despite the comprehensive scope outlined above, there are certain limitations that need to be acknowledged:

- The design of the board had to be completed in two and a half weeks, starting from March 10th to March 28th, 2025.
- The project is bound by a strict financial limit of 70 U.S. dollars, which requires optimal part selection and minimal waste. It covers the acquisition of all components necessary for the design, fabrication of the PCB, mechanical fabrication, and assembly.

1.3 GitHub Link

<https://github.com/M595-D0T/Micromouse-Project.git>

Chapter 2

Requirements Analysis

2.1 Requirements

The requirements for a micromouse power module are described in [Table 2.1](#).

Table 2.1: User and functional requirements of the power subsystem.

Requirement ID	Description
R01	All ICs must fit onto PCB
R02	PCB must fit onto the MM
R03	PCB should support both slow and fast charging modes
R04	Meet the budget requirements WITHOUT the use of additional resources
R05	Battery must charge when power is supplied
R06	Use a high-side driving configuration
R07	The system must allow bidirectional control of up to 4 motors (2 main and 2 auxiliary) with adjustable speeds

2.2 Specifications

The specifications, refined from the requirements in [Table 2.1](#), for the micromouse power module are described in [Table 2.2](#).

Table 2.2: Specifications of the power subsystem derived from the requirements in [Table 2.1](#).

Specification ID	Description
SP01	Charge battery from the 9V input pin
SP02	Slow and fast charging current for the battery must be 200mA, and approximately 600mA $\pm$ 100mA respectively
SP03	Battery current or voltage shall be monitored via INA219 on I2C, with addresses configured per hardware documentation
SP04	2x out of the 4x motors must each draw 200mA at the highest voltage of a 1S1P battery. The other 2x motors must draw 500mA
SP05	All ICs and supporting components must fit within a PCB area of 82mm $\times$ 60mm

2.3 Acceptance Testing Procedures

Table 2.3: Subsystem acceptance tests

ID	Description	Testing Procedure	Pass/Fail Criteria
AT01	Test that the dual charging mode circuit can correctly switch between 200mA and 600mA $\pm$ 100mA charging modes when the mode select circuit is toggled. This confirms that the current-limiting configuration behaves as expected.	-Use a jumper to toggle the FAST_CHARGE_CTRL pin to switch between current modes. Measure the charging current going into the battery using a multimeter.	Fail if slow charging current is not in the range of 200mA $\pm$ 10% and fast charging current is not in the range of 600mA $\pm$ 100.
AT02	Test if the INA219 is correctly configured on the I2C bus. This ensures I2C communication integrity.	-Verify I2C communication integrity by capturing I2C waveforms using an oscilloscope.	Pass if the waveform for low charging mode is a straight line at 0 V, and for high charging mode is a square wave with a max value of 3.3 V.
AT03	Check if the INA219 accurately measures battery voltage and current. This ensures reliable power monitoring functionality.	-Apply a load of 1 ohm to the load and measure current across the load using a multimeter.	Pass if the measurements return values of 200mA $\pm$ 10% and 600mA $\pm$ 100 correspond to those of the INA219.
AT04	Test that the motor driver output pins can bidirectionally control 4 DC motors; 2 at 200mA and 2 at 500mA using the assigned control pins and power from a 1S Li-ion battery. This ensures full compliance with system motor control requirements.	-Use a control jumper to connect a 3.7V DC source to the motor driver's direction control input pins. Apply a 1-ohm load to the output pins for the two motors rated at 200mA, and a 0.4-ohm load for the two motors rated at 500mA. Use a multimeter to directly measure the output current across each load to verify the current draw and motor direction.	The test is considered a pass if all motors spin, both 200mA motors draw current within the range of 200mA $\pm$ 10%, and both 500mA motors draw current within the range of 500mA $\pm$ 10%.
AT05	Check if the PCB dimensions are met. This ensures that all ICs and supporting components fit within the PCB.	-Physically measure PCB dimensions with a ruler.	Pass if measurements return a length of 82mm and a width of 60mm.

Chapter 3

Subsystem Design

3.1 Design Decisions

Design decision 1: Using an IC or a MOSFET for the motor driver circuit

Below is a table for the comparison of an IC and MOSFET considered during the design process in order to meet R07 and SP04:

Table 3.1: Comparison of Two Motor Driver Circuit Configurations

Feature	Dual H-Bridge IC(DRV8833PW)	Discrete H-Bridge using MOS-FET(IRF540N)
Motors Controlled	2 motors at once	4 x IRF540Ns per motor
Control	Bidirectional	Bidirectional
Complexity	Low	High
Current Handling	Up to 2A	33A
Size	Compact	Large
Price on JLCPCB	$0.69 \times 2 + 3 \text{ (extended)} = \4.38	$0.31 \times 16 + 3 \text{ (extended)} = \7.96

Final choice: For this part of the design, a Dual H-bridge IC was selected to provide full control over the motor's direction and braking, as required by R07 and SP04, with the DRV8833PW IC[1] allowing for control of 2x motors at once. While other options were considered (like using an IRF540N MOSFET), it did not allow for bidirectional control of the motors and cost a lot compared to the IC; much larger in size, which would not be practical for the MM.

Design decision 2: Integrated Circuit for dual charging modes circuit

R03 requires the battery to have two charging modes with the current for each mode specified by SP02. The following are some ICs considered for the design of the dual charging modes circuit, taking into account the charging current, voltage, availability, and price:

Table 3.2: Comparison of Different ICs Considered

IC	Charge Current	Charge Voltage	Availability On JLCPCB and Price
TP4056	Programmable Charge Current Up to 1000 mA	4.2V	Stock available 77520 at \$0.18 each
MCP73832	15 mA to 500 mA	4.20 V, 4.35 V, 4.40 V, 4.50 V	Stock available 107 at \$1.14 each
MAX1555	100mA(max)	0-7 V	Stock available 28 at \$1.55 each

Final choice: Three ICs were considered, all with similar electrical characteristics. The MAX1555 [2] was eliminated because its maximum charge current of 100mA did not meet SP02. Ultimately, the TP4056 [3] was selected over the MCP73832 [4] due to its suitable electrical parameters, greater availability, and low cost.

Design decision 3: INA219 Circuit Configuration

From R05, the battery must charge when power is supplied. To make sure of this, a battery monitor is needed. Hence, it was necessary to make a design decision about the circuit configuration of the INA219. The datasheet provides two circuit configurations[5]:

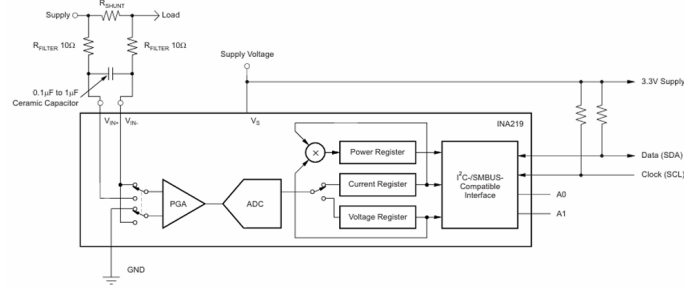


Figure 3.1: INA219 circuit configuration 1

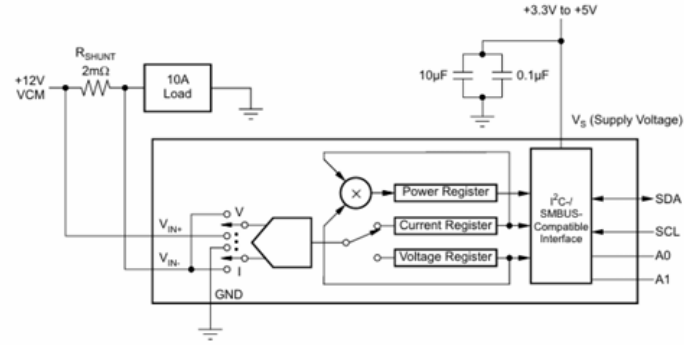


Figure 3.2: INA219 circuit configuration 2

Configuration 2 uses fewer components compared to Configuration 1, which meets R05. It also incorporates bypass capacitors that enhance voltage stability, which is essential for protecting sensitive circuit components. Configuration 1, on the other hand, includes two resistors (RF1 and RF2) and a capacitor, forming an input filter circuit. It also features two pull-up resistors to ensure proper functionality of the I2C communication interface.

Final choice: To optimize the design, a hybrid configuration was implemented by selectively retaining crucial components. The two pull-up resistors were kept to ensure stable logic states and avoid floating inputs. The bypassing capacitors were also kept as a safeguard to compensate for potential voltage regulator instability.

3.2 Final Design

Solution to Design decision 1: Using an IC or a MOSFET for the motor driver circuit

The final design uses two DRV8833PW ICs to control 4 bidirectional brushed DC motors using only 8 logic pins total (INA1, INA2, INB1, INB2, INC1, INC2, IND1, and IND2), while providing sufficient current (up to 2 A) per motor. Its integrated H-bridge design simplifies the circuit and meets the system's motor control requirements efficiently (R07 and SP04).

Solution to Design decision 2: Integrated Circuit for dual charging modes circuit

The TP4056 is used in the final design to manage the charging of the single-cell Li-ion battery. It supports switching between low and high current charge modes, enabling the system to charge at either 200 mA or approximately 600 mA, depending on the selected mode..

Solution to Design decision 3: INA219 Circuit Configuration

The final design includes a current monitor (INA219), which monitors the battery on the I2C bus. It is connected to bypass capacitors as a safety measure in case of a fault in the monitor, and pull-up resistors for proper I2C communication.

Below is a schematic of the final solution:

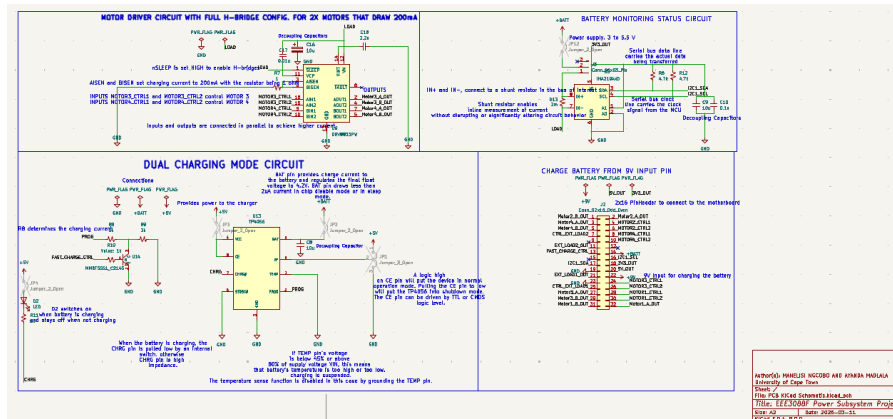
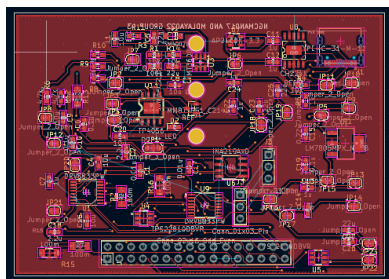
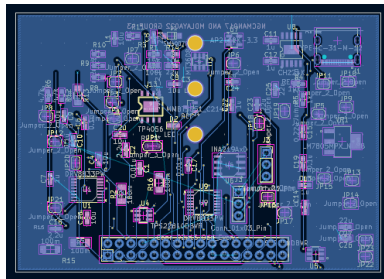


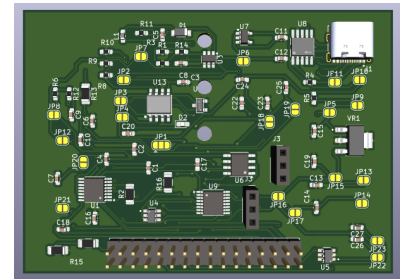
Figure 3.3: Schematic of the final solution



(a) Front PCB



(b) Back PCB



(c) 3D PCB

Figure 3.4: PCB

3.3 Failure Management

Table 3.3: Failure Management Processes Taken During The Design Process To Mitigate Failure

Name	Description
Open Jumper Isolation	Jumper headers are placed in series with critical signal paths such as the dual charging IC input(TP4056) and motor driver circuit input (DRV8833PW), allowing for the disconnect or rerouting of signals using jumper caps or wires. This enables isolation of faulty subsystems or rerouting to alternate pins if traces are incorrect.
Jumper Voltage Verification	Utilize jumper pins as convenient test points to measure voltage levels across critical power rails, such as the 9V input pin, where the battery charges. This allows for quick verification of whether power is correctly reaching different sections of the board.
Using smaller component size	The use of very small resistors and capacitors avoided to allow for desoldering if they malfunction, allowing component replacement to be possible.

Table 3.4: Bill of Materials (BOM) Categorized by JLCPCB Component Type (Basic vs Extended)

Component	Quantity	Individual Cost(USD)	Total Cost(USD)	Classification
DRV8833PW IC	2	0.69	4.38	Extended
TP4056 IC	1	0.18	3.18	Extended
INA219 IC	1	0.77	3.77	Extended
Load Resistor (4.7k Ohm)	2	0.01	0.02	Basic
Load Resistor (2m Ohm)	1	0.0582	3.0582	Extended
Load Resistor (100m Ohm)	4	0.01	0.04	Basic
Load Resistor (2k Ohm)	1	0.01	0.01	Basic
Load Resistor (3k Ohm)	1	0.01	0.01	Basic
Load Resistor (1k Ohm)	2	0.01	0.02	Basic
MMBT5551 Transistor	1	0.02	0.02	Basic
LED	1	0.01	0.01	Basic
Capacitor (10u)	4	0.01	0.04	Basic
Capacitor (0.1u)	1	0.01	0.01	Basic
Capacitor (0.01u)	2	0.01	0.02	Basic
Capacitor (2.2u)	2	0.01	0.02	Basic
TOTAL		1.8182	14.6082	

Note: Components selected from JLCPCB's Extended Library incur an additional handling fee of \$3.00 per unique part number, regardless of quantity. This is due to special sourcing and handling requirements.

3.4 System Integration and Interfacing

To integrate the subsystem with the rest of the system, the power subsystem makes important connections with the motherboard, the battery, and the four motors.

Table 3.5: Interfacing specifications

Interface	Description	Pins/Output
I001	Power Subsystem PCB to motherboard – Power distribution interface for delivery of power from power subsystem PCB to the motherboard.	- Power Subsystem MOTOR2_CTRL1 Pin 4 to Motherboard Pin PD12 - Power Subsystem MOTOR2_CTRL2 Pin 6 to Motherboard Pin PD13 - Power Subsystem MOTOR4_CTRL1 Pin 8 to Motherboard Pin PD14 - Power Subsystem MOTOR4_CTRL2 Pin 10 to Motherboard Pin PD15 - Power Subsystem MOTOR3_CTRL1 Pin 24 to Motherboard Pin PC6 - Power Subsystem MOTOR3_CTRL2 Pin 26 to Motherboard Pin PC7 - Power Subsystem MOTOR1_CTRL1 Pin 28 to Motherboard Pin PC8 - Power Subsystem MOTOR1_CTRL2 Pin 30 to Motherboard Pin PC9 - Power Subsystem FAST_CHARGE_CTRL Pin 13 to Motherboard Pin PB4 - Power Subsystem I2C1_SCL Pin 16 to Motherboard Pin PB8 - Power Subsystem I2C1_SDA Pin 17 to Motherboard Pin PB9 - Power subsystem High Voltage (9V) Pin 22 to Motherboard HV - Power Subsystem CTRL_EXT_LOAD1 Pin 25 to Motherboard Pin PD11 - Power Subsystem CTRL_EXT_LOAD2 to Motherboard Pin PD2 - GND: Pins 19 and 23 to GND
I002	Power subsystem to bidirectional motors – High-current (2 A max) interface supplying PWM-controlled power to motors.	- MOTOR2_B_OUT and MOTOR2_A_OUT to Motor 2 - MOTOR4_A_OUT and MOTOR4_B_OUT to Motor 4 - MOTOR1_B_OUT and MOTOR1_A_OUT to Motor 1 - MOTOR3_B_OUT and MOTOR3_A_OUT to Motor 3
I003	3.7V Battery to Power subsystem PCB – Input connection for the Li-ion battery with charge management.	- BATT to + terminal - GND to – terminal
I004	USB Type C to Power subsystem PCB – 5V DC input for charging and system power, compliant with USB-PD.	Used for power delivery, data transfer, and microcontroller programming.
I005	3V3 and 5V Out from Power subsystem PCB to Motherboard – Regulated output lines for peripherals.	3V3 and 5V Out Pin 18 and Pin 20 to Motherboard 3V3 and 5V Out

The diagram below illustrates the interface of the power subsystem with its key components.

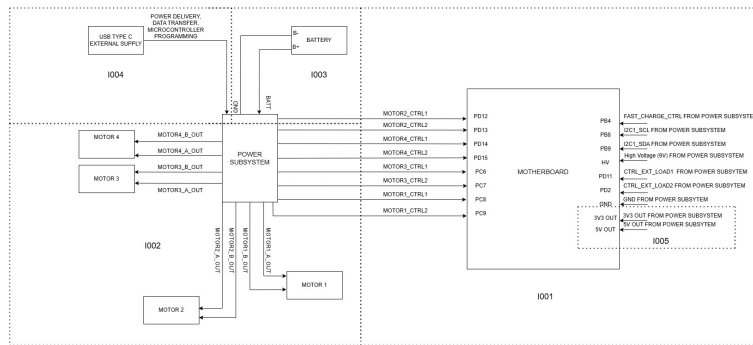


Figure 3.5: System interfacing diagram

Chapter 4

Discussion and Results

4.1 Critical Analysis

Analysis of AT01: Dual Charging Mode Current-Limiting Verification

Table 4.1: AT01 Expected and Actual Results

What the ATP Tests	Expected Result(s)	Actual Result(s)
ATP01 tests whether the dual charging mode circuit can correctly switch between 200 mA and 600 mA ± 100 mA.	- When the charging mode circuit is toggled to low mode, the measured charging current should be 200 mA $\pm 10\%$. - When toggled to high charging mode, the measured current should be 600 mA ± 100 mA.	0 mA 0 mA

The charging circuit failed to regulate current, outputting 0 mA in both modes (expected: 200 mA in low mode, 600 mA in high mode). This prevented battery charging and posed overcharge risks. A 5 V DC power supply, verified by a multimeter, is supplied to the input pin of the circuit. A load of 25 ohms is applied to the output pin, and the output voltage is measured using a multimeter, which outputs 0 V for both modes; hence, no current is drawn. A closer inspection of the board on KiCad shows that the autorouting tool did not complete routing all the traces to the charging circuit due to components (U13, C2, and D2) being tightly spaced together as shown in Figure 4.1, thus some parts of the circuit were not connected. This ATP failed.

Analysis of AT02 and AT03: Battery Monitoring Circuit

Table 4.2: AT02 and AT03 Expected and Actual Results

What the ATP tests	Expected Result(s)	Actual Results(s)
The tests validate I2C connectivity and measurement accuracy of the INA219 for reliable battery monitoring.	The INA219 must respond to its configured I2C address (ensuring A0/A1 pins are not both grounded) and allow read/write operations and accurately measure the 1S1P battery voltage (typically 3.7 V nominal, 4.2 V fully charged) within $\pm 5\%$ error.	No communication with I2C and no measured voltage drop across the INA219s load.

Initially, the ATPs for these tests were different. After conducting thorough research, the conclusion was that code is needed to get the INA219's voltage and current register values, and to check if the I2C communication works. Find the new ATP at the end of this section Table 4.5 The STM returned a value of 0 for both the current and voltage values across the INA219's shunt resistor from their respective addresses. The root cause of this failure is that all the pins of the INA219 were not connected due to incomplete routing. This means that the charging status of the battery cannot be monitored, which might lead to overcharging, undercharging, or difficulty in diagnosing power-related issues during operation. This ATP failed.

Analysis of AT04: Motor Driver Output Capability

Table 4.3: AT04 Expected and Actual Results

What the ATP Tests	Expected Result(s)	Actual Result(s)
Tests bidirectional control of 4 motors: 2 at 200 mA, 2 at 500 mA	- Motors 1 and 2: 200 mA each - Motors 3 and 4: 500 mA each	- Motors 1 and 2: 0 mA, 0 mA - Motors 3 and 4: 0 mA, 0 mA

The motor driver circuits did not function as intended. A DC power supply of 3.7 V, verified by the multimeter, is applied to the control pins of Motors 1, 2, 3, and 4 in the order specified. A load of 1 Ohm is applied to the output pins in order to measure the current across it. The multimeter, set to the 200 mA range, registered a value of 0 mA for all the motors, in the order specified. Since these motors drive the actuation for the MM, the failure directly compromises all mobile operations. Inspection of the PCB layout on KiCad revealed that the autoroute tool partially routed the tracks to the two motor drivers. This highlights a flaw in the design verification process and emphasizes the need for manual review in future designs. Given that the output current is 0 mA due to routing errors, this ATP failed.

Analysis of ATP05: PCB Dimensional Accuracy

Table 4.4: AT05 Expected and Actual Results

What the ATP tests	Expected Result(s)	Actual Results(s)
AT05 tests whether the PCB's dimensions are met.	The expected length and width must be a maximum of 79.5 mm and 59.5 mm, respectively.	Physical measurements of the board's dimensions returned a length of 79 mm and a width of 59 mm.

The results obtained for the length and width of the PCB were within the range of the expected results, taking into consideration that the manufacturer of the board, JLCPCB, has a very low dimensional tolerance of ± 0.2 mm. Since the actual dimensions of the PCB are a bit less than the maximum stipulated dimensions, it ensures that the PCB fits onto the MM. Failure to meet these dimensions would have resulted in the PCB not fitting into the designated enclosure. Proper use of design software (KiCad) with built-in Design Rule Checks (DRC) to flag errors before fabrication ensured that the

requirements and specifications associated with this ATP were met. The 0.63% length and width reduction suggests that there could have been a copper etching effect, causing the board to shrink. This ATP passed. See Figure 4.2:

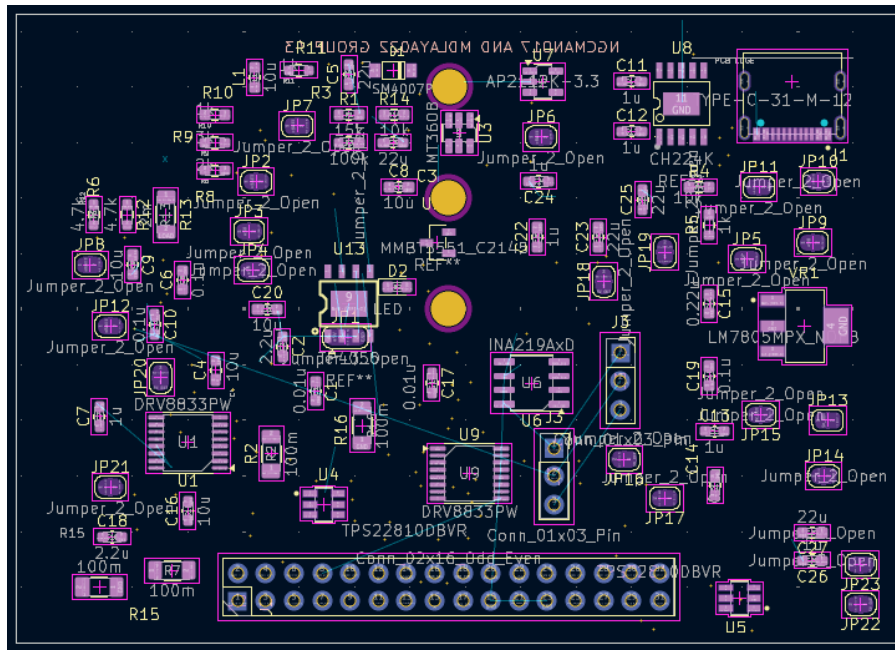


Figure 4.1: PCB layout on KiCad showing unrouted traces



Figure 4.2: Measurements for board dimensions

Table 4.5: Revised ATP for ATP02 and ATP03

Category	Task	Done
Hardware Setup	Connect STM32F0 GND to INA219 GND	☐
	Wire PB6 (SCL) to INA219 SCL	☐
	Wire PB7 (SDA) to INA219 SDA	☐
Software Configuration	Configure PB6 (SCL) and PB7 (SDA) as I2C pins in firmware	☐
	Initialize I2C communication (HAL/LL libraries)	☐
I2C Bus Scan	Scan bus to confirm INA219 detection (address 0x40)	☐
INA219 Initialization	(Optional) Configure INA219 calibration registers	☐
Data Reading	Read shunt voltage	☐
	Read bus voltage	☐
	Read calculated current	☐
Validation	Compare readings with multimeter (max 5% deviation)	☐
	Repeat tests for consistency	☐

Chapter 5

Conclusion

This report has focused on the essential tasks involved in designing, developing, manufacturing, and testing the power subsystem. The primary role of the subsystem, which is to provide the MM robot with the necessary power to operate effectively and solve mazes autonomously, is not achieved. Critical failures were identified during testing, primarily related to the charging circuit, battery monitoring circuit and the motor driver circuit. These outcomes indicate that the design did not meet critical functional specifications, particularly in relation to motion control and battery monitoring, and charging. Despite the lack of successful operation, the testing process exposed crucial weaknesses in the design workflow, most notably, the limitations of relying on autorouting without manual inspection, and insufficient design verification prior to fabrication. These insights are valuable for refining future iterations of the system.

5.1 Recommendations

To address the identified failures and enhance subsystem performance, the following improvements are proposed:

- Use JLCPCB's Gerber viewer to verify layer alignment and tolerances before fabrication.
- Perform manual routing for traces to prevent autorouter failures.
- Implementing a modular board layout could help with debugging, should any part of the system be faulty, and isolating that part from the rest of the system.
- Thoroughly perform design rules check (DRC) and electrical rules check before finalising the PCB layout.

Project Reflection

Design Review

The project did not meet its intended outcomes but offered key insights at each stage:

- **Design:** Insufficient simulation and datasheet analysis led to early design flaws.
- **Layout:** Autorouting caused critical errors. Manual routing and full DRC/ERC checks are essential.
- **JLCPCB Files:** Gerbers should be verified in a viewer, with netlist and continuity checks pre-submission.
- **Testing:** Whole-system testing made debugging hard. A modular testing approach is better.

Post-Facto Gantt Chart

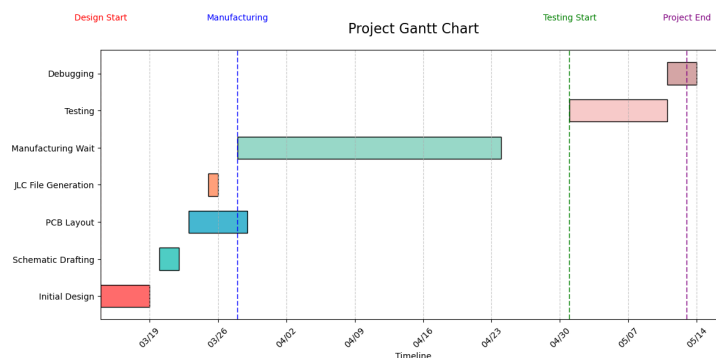


Figure 5.1: Post-facto Gantt chart with task dependencies

Reducing Design Cost

- Replace discrete components with integrated ICs.
- Use single-layer PCBs where possible.
- Avoid uncommon or costly components.

Unequal Motor Parameters

If motor characteristics vary, use encoder feedback and PID control to adjust PWM and maintain straight-line motion.

Motor Stall or Short-Circuit

Stalls or shorts cause current spikes that may damage hardware. To mitigate:

- Add current-limiting resistors or fuses.
- Use drivers with overcurrent or thermal protection, monitor current in firmware, and shut down if limits are exceeded.

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